

52

SARDAR PATEL UNIVERSITY

B.Sc. (CA & IT) / M.Sc. (IT) (Int.) (IV Semester) Examination

Tuesday, 10th April 2018

10.00 am - 1.00 pm

PS04FIIT01 : Computer based Numerical and Statistical Method

Total Marks: 70

Q-1 Write the correct option in the answer book.**[10]**

- (1) If $f(a) < 0$, $f(b) > 0$ and if $x_0 \in (a, b)$ is first approximation with $f(x_0) < 0$ then in bisection method _____
 (a) x_0 is to be replaced by a (b) a is to be replaced by c
 (c) b is to be replaced by x_0 (d) x_0 is to be replaced by b
- (2) The number 0.01850×10^3 has _____ significant digits
 (a) 3 (b) 4 (c) 5 (d) 6
- (3) All the formulae of interpolation are based on the fundamental assumption that the given data can be expressed as a _____.
 (a) Polynomial (b) Equation (c) Algorithm (d) None
- (4) _____ Method is used if the estimated value lies towards the end of the difference table.
 (a) Divided difference (b) Forward difference
 (c) Backward difference (d) None of the above
- (5) y depends on x can be written as _____
 (a) $f(x)$ or y_x (b) $f(xy)$ (c) $f(yx)$ (d) none
- (6) _____ is called the backward difference operator.
 (a) Δ (b) ∇ (c) $\emptyset$ (d) $\cup$
- (7) We can find solution of system of linear, algebraic equations using _____
 (a) Newton-Raphson method (b) Bisection method
 (c) Gauss-Seidel method (d) None of these
- (8) The system of linear equation $AX = B$ can be solved by matrix inversion method only if _____
 (a) $A \neq 0$ (b) $|A| \neq 0$ (c) $|A| = 0$ (d) A is symmetric
- (9) Forecasts _____
 (a) become more accurate with longer time horizons
 (b) are rarely perfect
 (c) are more accurate for individual items than for groups of items
 (d) all of the above
- (10) Gradual, long-term movement in time-series data is called _____.
 (a) seasonal variation (b) cycles
 (c) trends (d) exponential variation

Q:2 Answer the following in short. (Any Ten)**[20]**

- (1) Write algorithm for bisection method.

(P.T.O.)

- (2) Define Relative error and absolute error.
- (3) Describe the stopping rules to obtain approximate solution for given non-linear equations.
- (4) Use the secant method to obtain approximate solution of the equation $x^3 - 5x - 3 = 0$. [initial approx. 2 & 3]
- (5) Define Interpolation.
- (6) Explain central difference table.
- (7) What is Time Series?
- (8) List only various direct and iterative methods.
- (9) If x lies in the upper half of the table and if $x = x_k$, then what is $\frac{dy(x)}{dx}$ and $\frac{d^2y(x)}{dx^2}$?
- (10) List the component of Time series.
- (11) What do you mean by Secular trend?
- (12) What do you mean by random or irregular Variation?

Q-3

- (a) Find the root of equation $x^3 - 4x - 9 = 0$ correct up to four decimal places using False- Position Method. [05]
- (b) Find the root of equation $x^3 - x^2 - 1 = 0$ correct up to four decimal places using Iterative method. (Take $a = 1.5$) [05]

Q-3

OR

- (c) Find the root of equation $x^3 - x - 1 = 0$ correct up to four decimal places using Bisection method. [05]
- (d) Find the root of equation $xe^x - 1 = 0$ correct up to four decimal places using Secant method. [05]

Q-4

- (a) Find the polynomial function $f(x)$ given that $f(0) = 2$, $f(1) = 2$, $f(2) = 12$ and $f(3) = 35$ find $f(5)$. [05]
- (b) Estimate by Newton's method of interpolation, the expectation of life at age 32 from the following table. [05]

Age (In Years)	10	15	20	25	30	35
Expectation of life (In Year)	35.3	32.4	29.2	26.1	23.2	20.5

Q-4

OR

- (c) The following table gives the census population of a town for the years 1931 to 1971. Estimate the population for the year 1935 and year 1965 by using an appropriate interpolation formula. [10]

Year	1931	1941	1951	1961	1971
Population	46	66	81	93	101

Q-5

- (a) Solve the following system of equations using Gauss-Seidel method.

$$10x_1 + x_2 + 2x_3 = 44$$

$$2x_1 + 10x_2 + x_3 = 51$$

$$x_1 + 2x_2 + 10x_3 = 61$$

[05]

- (b) Write the comparison between direct and iterative methods for solution of system of linear equations. [05]

Q-5

OR

- (c) The distance (s) covered by a car in given time (t) is given in the following table:

Time(minutes)	10	12	14	16	18
Distance(km)	12	15	20	27	37

[05]

Determine the speed of the car at $t=13$ minutes.

- (d) Use Backward difference formula to compute $\frac{dy}{dx}$ at $x=7$ and $x=7.5$,
From the following table:

x	3	4	5	6	7	8
y	4	6.6	7.7	9.0	10.5	12.2

[05]

- Q.6 Calculate the trend values by the method of moving average, assuming a four-yearly cycle from the following data relating to sugar production in India.

YEAR	SUGAR PRODUCTION (lakh tones)	YEAR	SUGAR PRODUCTION (lakh tones)
1971	37.4	1977	48.4
1972	31.1	1978	64.4
1973	38.7	1979	58.4
1974	39.5	1980	38.6
1975	47.9	1981	51.4
1976	42.6	1982	84.4

[06]

(C.P.T.O.)

- (b) Obtain the trend from the time series given below by method of moving average of 4 years

Year	1958	1959	1960	1961	1962	1963	1964	1965	1966	1967
Y	50.0	36.5	43.0	44.5	38.9	38.1	32.6	41.7	41.1	33.8

[04]

OR

- Q.6 Obtain the trend from the time series given below by method of moving average of (i) 3 years (ii) 5 years

[06]

Year	1961	1962	1963	1964	1965	1966	1967	1968	1969	1970
Y	500	540	550	530	520	560	600	640	620	640

- (d) What is time series? List various components of time series and explain one of them.

[04]